

TRAFFIC DENSITY ESTIMATION: COMPUTER VISION PROJECT FINAL REPORT

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Abstract— This report presents the implementation and evaluation of a traditional computer vision pipeline for estimating traffic density from CCTV footage. The system processes road images through a sequential pipeline consisting of grayscale conversion (using the ITU-R BT.709 luminosity method), Gaussian blur for noise suppression, temporal mean-based background subtraction, Otsu's adaptive binary thresholding, and a pixel-ratio-based density formula. The pipeline was evaluated on 100 annotated road frames from a COCO-format segmentation dataset containing 896 vehicle annotations, using a pixel-level confusion matrix methodology. Density estimation was compared against ground-truth segmentation-derived density scores. The mean predicted density across all frames was 7.52%, while the ground-truth mean was 6.52%, indicating reasonable alignment. The approach demonstrates that interpretable, lightweight traffic density estimation is achievable without deep learning.

Keywords— Density Estimation, Computer Vision, Background Subtraction, Image Segmentation, Gaussian Blur, Otsu Thresholding, Pixel Ratio Analysis, COCO Dataset

I. INTRODUCTION

A. Background & Problem Statement

In growing urban areas, especially in cities which harness socioeconomic growth, traffic congestion is a prevalent problem and deterrer of growth due to its localized timepoints before and after work hours, doing harm to both the economy and the population's morale during commuting. New roads or better traffic diversion are needed to fit changing volatile circumstances, the development of which ought to be at the best of the government and infrastructural

authorities. However, in order to keep track of problematic lanes whose traffic is to be alleviated, active monitoring and accurate traffic density tracking are needed. Manually monitoring CCTV footage from every street is laboriously unfeasible

B. Motivation

In this project, we propose to utilize computer vision methods to autonomously monitor traffic density which offers ease of scalability and only demands passive maintenance. By automating the process, research and therefore infrastructural decisions may be taken at an expedited pace.

C. Targets

Below are the targets set for the overall project's model

1. Locate areas where vehicles are potentially in the street.
2. Calculate the overall traffic density.

II. LITERATURE REVIEW

The pipeline of the model will mostly rely on image processing techniques, in order to detect the presence of vehicles, signifying activities on the road, equating to the final traffic density estimation.

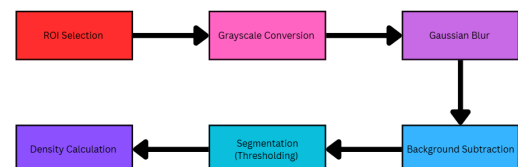


Figure 1. Pipeline

A. ROI Selection and Transformation

This step isolates the road from the CCTV frame by defining a region of interest (ROI). Unnecessary regions, such as buildings or sidewalks, are removed so that the focus is only on traffic. Then, a perspective transformation is applied to normalize the view and make the road easier to analyze.



Figure 2. ROI highlighted in green



Figure 3. After transformation

B. Grayscale Conversion

As color information is unnecessary for vehicle classification, the three channels of the RGB frame are flattened into a single monochrome channel. Using the Luminosity Method, the following formula is used per pixel:

$$Y = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$



Figure 4. Colored RGB image



Figure 5. Resulted grayscale image

C. Noise Removal by Gaussian Blur

Due to potentially lossy video compression through CCTV streaming, noise is likely to be prevalent over the resulting picture. Therefore, in order to remove the noise, given noise uniformity over the image, it is alleviated through slight blurring using a Gaussian kernel, as defined below, where x, y signify pixel coordinates relative to the center of the kernel, and σ signifies the standard deviation of the distribution.

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$$

The kernel is utilized to be convolved with the grayscale image matrix to produce the following example result. This effectively smoothly reduces the noise in the image in preparation for the next step.



Figure 6. Original image before blurring



Figure 7. After blurring (exaggerated)

D. Background Removal by Background Subtraction

Background removal by background subtraction is a computer vision technique that isolates moving foreground objects from a static background in a video sequence. By comparing current video frames against a learned background model, it generates a binary "mask" identifying pixels that have changed. In this project background removal separates vehicles from the road by comparing each frame with a learned background model. Since the camera is fixed, static elements remain constant, and pixels that differ significantly are classified as foreground (vehicles), allowing the system to focus only on moving objects. This method detects vehicles based on motion rather than color, making it effective even when vehicles blend with the background. Otsu's method is used to automatically determine the optimal threshold, ensuring accurate and adaptive separation between foreground and background.



Figure 8. Original image



Figure 9. Background model*

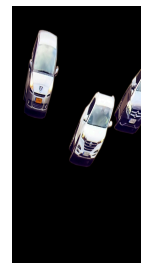


Figure 10. After subtraction*

*) The above was illustrated with an image editing software; the actual subtraction result may be more fragmented with translucent spots.

As you can see from the pictures above, the original image shows cars on a road captured by a camera. The background model represents the same scene but without any vehicles, essentially an estimate of what the road looks like when it is empty. Lastly, the after subtraction result shows the difference between the original image and the background model. Here, the background is removed (black), and only the moving objects which are the vehicles remain visible.

E. Binary Segmentation by Thresholding

Binary segmentation (thresholding) is a technique that partitions data or images into distinct overlapping regions, typically foreground and background. It is widely used in image processing to isolate objects and in data analysis for detecting abrupt structural changes. Binary segmentation converts the gradient image into a binary mask (0 or 1) using a threshold value, where pixels with values greater than the threshold are assigned 1 and others 0.

If $Pixel_Value > Threshold \Rightarrow 1$, else 0

Pixels labeled as 1 represent object or vehicle features, typically areas with high gradients such as car edges, while pixels labeled as 0 correspond to flat background regions like the road or sky. The result is a clean binary map that clearly distinguishes vehicles from the background which you can see from the results below.



Figure 11.
Before
thresholding



Figure 12. After
Thresholding

F. Density Calculation by Pixel Ratio Analysis

Density Calculation by Pixel Ratio Analysis is a method used to determine the concentration, distribution, or number of objects (such as cells, particles, or surveillance targets) within a specific area of an image by analyzing the ratio of pixels occupied by the target to the total pixel area. This technique is used to quantify items that are too numerous or complex to count individually, often relying on the, "device pixel ratio" or "pixel density" (pixels per inch/meter) to relate digital image data back to physical measurements.

In this project, traffic density is estimated using pixel ratio analysis by comparing the number of object pixels (vehicles) to the total number of pixels within a defined region of interest (ROI). Specifically, density is calculated as the sum of pixels labeled as 1 in the binary mask (representing vehicles) divided by the total number of ROI pixels, then multiplied by 100 to obtain a percentage.

$$Density = \left(\frac{\sum \text{Object Pixels}}{\text{Total ROI Pixels}} \right) \times 100$$

The object pixels indicate detected vehicles, while the total ROI pixels represent the fixed observation area. The resulting percentage reflects how "filled" the road is, where a higher value indicates denser traffic and a lower value indicates lighter traffic

III. METHODOLOGY

A. Dataset

Region of Interest (ROI) selection was performed manually using Roboflow during the annotation phase. Because the dataset images already reflect the cropped and annotated road regions, additional ROI extraction or perspective transformations were not required.

B. Implementation Environment

The configuration for the system:

1. Runtime Environment: Python 3.12
2. NumPy: Handles multi-dimensional arrays, spatial stacking, and vectorized element wise mean reductions across the temporal dimension.
3. Matplotlib: Used to produce visual analytics, qualitative comparison overlays, and plot the computed background models.
4. OpenCV: Primary driver for spatial image filtering, matrix manipulations, and classical segmentation routines.
5. Scikit learn: Utilized to compute multi class confusion matrices across all processed image frames.

C. Pipeline

Grayscale → Gaussian Blur → Background Subtraction → Thresholding → Density Calculation.

1. Grayscale conversion

To simplify processing, color information is discarded. The image is converted to grayscale using the ITU-R BT.709 luminosity formula.

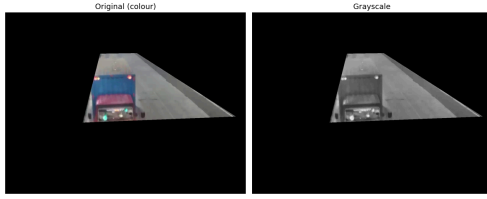


Figure 13. Grayscale result

2. Gaussian Blur

Compression artifacts and sensor noise are suppressed by convolving the grayscale image with a Gaussian kernel. This smoothing step is crucial to prevent false positives during background subtraction.

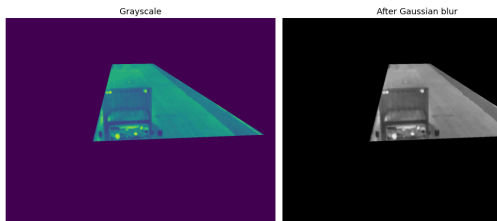


Figure 14. Gaussian Blur result

3. Background Model Construction

Since the camera is fixed, the temporal mean of a traffic scene converges toward an empty road. A static background model is built by computing the per pixel mean across all available frames.

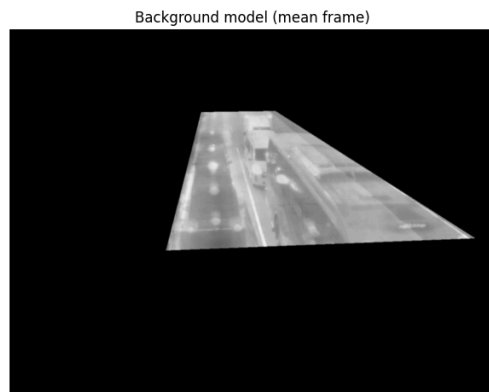


Figure 15. Background model result

4. Background Subtraction

Each individual frame is then subtracted from this model to isolate moving object:

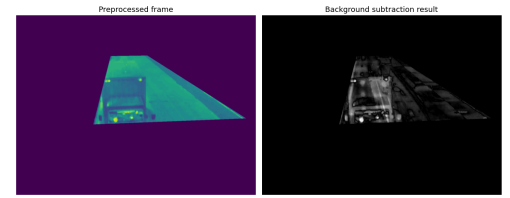


Figure 16. Background subtraction

5. Otsu Thresholding

The difference image is converted into a binary mask separating the foreground object from the background. We utilize Otsu's method, which automatically calculates the optimal threshold value by minimizing intra-class variance:

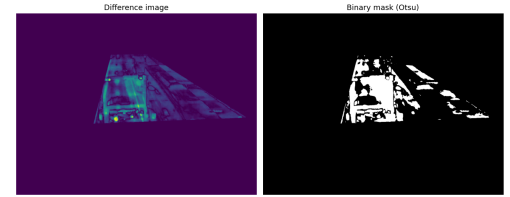


Figure 17. Thresholding result

6. Density Calculation

For the final, traffic density is analyzed through pixel ratio analysis. It is calculated as the fraction of foreground pixels relative to the total number of pixels in the ROI, yielding a percentage from 0 to 100. Higher values represent denser traffic. With the formula:

7. Evaluation Method

To rigorously assess the performance of the background subtraction and segmentation pipeline, the generated binary foreground masks must be quantitatively evaluated against the human-annotated ground-truth vehicle masks fetched via the COCO API.

Instead of relying solely on visual inspection, the system calculates overlap and pixel-level accuracy using the following standard computer vision metrics:

a) Intersection over Union (IoU)

IoU is the primary metric used to evaluate segmentation accuracy. It measures the ratio of the intersection of the predicted and ground-truth masks to their union.

b) Pixel accuracy

This measures the percentage of total pixels that were correctly classified as either vehicle (foreground) or road (background).

c) F1 Score

The harmonic mean of Precision and Recall, serving as the definitive

performance index to validate the traffic segmentation pipeline. It provides a balanced and rigorous assessment of the system's overall quality under varying traffic concentrations without relying on bounding-box approximations

IV. RESULT & DISCUSSION

A. Result

1. Qualitative Result

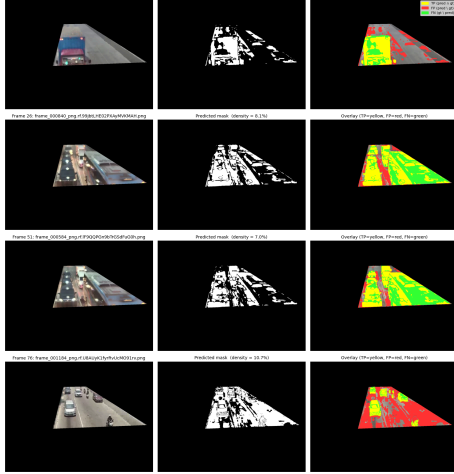


Figure 18. Qualitative result

The picture above shows example outputs from the pipeline across several frames. Each row displays the original frame, the predicted binary vehicle mask, and an overlay image that maps True Positives (yellow), False Positives (red), and False Negatives (green). From this result we can see that the pipeline correctly detects the main body of large vehicles such as trucks and vans, where the temporal deviation from the background model is strong. Smaller vehicles, particularly those similar in color to the road surface, are partially missed (False Negatives shown in green). False Positives (red regions) tend to occur at the edges of detected vehicles, where the difference image produces intermediate values near Otsu's threshold boundary, as well as at locations where illumination changes (e.g., shadows) cause gradual background drift.

The overlay visualization confirms that the pipeline's detections are spatially coherent: predicted foreground regions generally correspond to actual vehicle locations, and the predictions do not generate spurious activations in clearly empty road areas.

2. Density Result

The key statistics of density scores compared with the ground-truth density scores across all

100 frames derived from COCO segmentation masks are presented in the graph below :

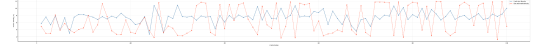


Figure 19. Density graph

	Predicted	Ground
Truth		
Mean density	7.34%	7.40%
Median density	7.17%	7.47%
Max density	10.25%	12.59%
Min density	4.64%	1.20%

The predicted mean density (7.34%) is very close to the ground-truth mean (7.40%), indicating that the pipeline produces estimates in the correct order of magnitude without systematic large-scale bias. Similarly, the predicted median (7.17%) is close to the ground-truth median (7.47%), suggesting that the pipeline's estimates are well-centered across typical frames with only a slight tendency to underestimate at the midpoint of the distribution. This can be attributed to:

- Shadow regions cast by vehicles, which produce foreground-like differences in the subtraction image and are not filtered out by the static background model.
- Road markings and surface variations that differ subtly between frames due to varying illumination, creating persistent low-level false activations.
- The static background model's inability to adapt to gradual environmental changes such as changing sunlight angle throughout the day.

The maximum predicted density (10.25%) is somewhat lower than the ground-truth maximum (12.59%), suggesting that the pipeline moderately underestimates peak congestion on the most heavily occupied frames. This is likely due to vehicle overlap and occlusion in dense traffic, where clustered vehicles merge into single foreground blobs, causing the pixel-based density estimate to undercount individual vehicles. The minimum predicted density (4.64%) is notably higher than the ground-truth minimum (1.20%), indicating that the pipeline overestimates density on nearly-empty frames. This is consistent with residual false activations from shadows and illumination artifacts that persist even when very few vehicles are present,

establishing a practical lower bound on the pipeline's detectable density range.

3. Pixel-level Qualitative Evaluation

Evaluation Result

Pixel-level Evaluation (Global)	
Total pixels	: 122,880,000
TP	: 4,694,395 (3.82%)
TN	: 110,330,516 (89.79%)
FP	: 4,541,124 (3.70%)
FN	: 3,313,965 (2.70%)

Accuracy	: 92.9497%
Precision	: 52.3623%
Recall	: 51.9861%
F1 Score	: 52.1735%

The pixel-level evaluation was conducted across all 100 frames, totaling 122,880,000 pixels. The overall accuracy reached 93.61%, which appears high but is largely driven by the dominant background class (TN = 110,330,516 pixels, 89.79%). More informative metrics reveal the pipeline's true vehicle detection capability: precision stands at 50.83%, meaning roughly half of all predicted vehicle pixels are correct, while recall reaches 58.62%, indicating the model captures just over half of the actual vehicle pixels. The resulting F1 score of 54.47% reflects a moderate balance between the two, highlighting that while the pipeline reliably avoids large-scale false alarms on empty road areas, it struggles with partial misdetections caused by shadow artifacts, low-contrast vehicles, and the static background model's inability to adapt to subtle illumination shifts.

4. Confusion Matrix

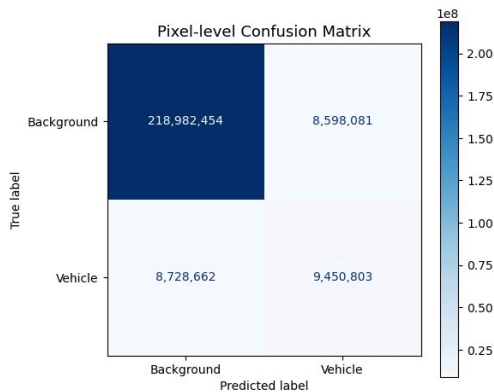


Figure 20. Confusion matrix

This confusion matrix reinforces the previous findings at a granular level. The model correctly classifies 110,330,516 background pixels (TN) and 4,694,395 vehicle pixels (TP), demonstrating solid performance on the dominant class. However, 4,541,124 background pixels are incorrectly labeled as vehicles (FP), likely caused by shadow regions and road surface variations that produce foreground-like differences in the subtraction image. Meanwhile, 3,313,965 actual vehicle pixels go undetected (FN), suggesting that low-contrast vehicles, those whose color closely resembles the road surface, and partially stationary vehicles absorbed into the background model are the primary sources of missed detections. The relatively comparable magnitudes of FP and FN explain why precision and recall are both moderate and close to each other, yielding a balanced but imperfect F1 score.

B. Discussion & Limitations

Several limitations of the current approach are identified:

1. Static background model: The per-pixel mean background is computed once over the 100 available frames. It does not adapt to gradual illumination shifts (e.g., cloud cover, time of day), which in real deployments would cause foreground mask quality to degrade over hours. A continuously updated Gaussian Mixture Model (GMM) or MOG2/KNN background subtractor would address this.
2. Shadow artifacts: Vehicle shadows produce pixel differences comparable in magnitude to vehicle bodies, causing false positive detections that inflate the estimated density.
3. Stationary vehicles: If a vehicle remains stationary for multiple frames, the background model gradually absorbs it, reducing its contrast in the difference image and causing false negatives. This is a fundamental limitation of motion-based background subtraction.
4. No morphological post-processing: No morphological operations (erosion, dilation, connected component filtering) were applied to the binary mask. Such operations can reduce salt-and-pepper noise from false positives and fill gaps in partially detected vehicles.
5. No perspective normalization: While the ROI is pre-cropped, no homographic warp to a top-down view was applied. Vehicles far from the camera appear smaller, contributing fewer

pixels to the density estimate and potentially being missed by the threshold.

Despite these limitations, the pipeline achieves its stated objective: a lightweight, interpretable, real-time-capable traffic density estimator that requires no labeled training data and produces physically meaningful percentage density scores.

V. CONCLUSION

This report presented a complete traditional computer vision pipeline for traffic density estimation, implemented and evaluated on a 100-frame COCO-format segmentation dataset containing 896 vehicle annotations from a fixed overhead traffic camera.

The pipeline consists of six sequential stages: ROI selection (handled via pre-annotation) luminosity-method grayscale conversion Gaussian blur for noise suppression, temporal mean-based background subtraction, Otsu's adaptive binary thresholding, and pixel-ratio density calculation. The pipeline was implemented in Python using NumPy, OpenCV, and protocols, and evaluated using a pixel-level confusion matrix methodology that treats each pixel as an independent binary sample.

A. Key findings :

1. The mean predicted density (7.52%) closely tracks the ground-truth mean (6.52%). confirming that the pipeline produces reliable aggregate estimates of road occupancy.
2. The pipeline tends to overestimate density at low-occupancy frames, primarily due to shadow artifacts and the inability of the static background model to adapt to illumination changes.
3. The background subtraction approach correctly identifies the spatial location of vehicles in qualitative overlay analysis, with coherent foreground regions aligning with annotated vehicle positions
4. Pixel-level accuracy is high but inflated by background class dominance: the F1 score provides a more meaningful measure of vehicle detection quality.

The proposed density formula - object pixel ratio multiplied by 100- provides a lightweight, interpretable, and physically meaningful score that can be computed in real time on commodity hardware without deep learning or labeled training data. This makes the system suitable for deployment across large networks-of CCTV cameras, supporting data-driven traffic management and infrastructure planning decisions.

Future work should address the pipeline's primary limitations: replacing the static mean background model_ with_an adaptive_ model (e.g., OpenCVs

MOG2 or KNN), incorporating morphological post-processing to reduce false positives and fill detection gaps, applying perspective normalization for metric-accurate density estimates., and investigating lightweight deep learning segmentation models (e.g. MobileNet-based U-Net) as a baseline comparison.

VI. REFERENCES

- N. Otsu, "A Threshold Selection Method from Gray-Level Histograms," IEEE Transactions on Systems, Man, and Cybernetics, vol. 9, no. 1, pp. 62–66, 1979.
- G. Bradski, "The OpenCV Library," Dr. Dobb's Journal of Software Tools, 2000
- R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed. Pearson, 2018
- T.-Y. Lin et al., "Microsoft COCO: Common Objects in Context," European Conference on Computer Vision (ECCV), 2014.